

ARC-AGI Project Report: CS7637 Knowledge-Based AI

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Abstract—This report presents the design, development, and evaluation of an ARC-AGI agent built to solve 96 abstract reasoning problems. It explains the agent’s core functionality, evaluates its performance across all problem sets, and examines both successful and unsuccessful cases to highlight its strengths and limitations. The report concludes by comparing the agent’s reasoning to human problem solving and reflecting on how the final system evolved over the course of the project.

1 INTRODUCTION

In 2019, French software engineer and artificial intelligence researcher François Chollet published a paper titled *On the Measure of Intelligence*. In this paper, Chollet examines intelligence through the lens of Algorithmic Information Theory and formally defines it as “the measure of its skill-acquisition efficiency over a scope of tasks, with respect to priors, experience, and generalization difficulty” (Chollet, 2019). More simply, intelligence can be understood as how quickly one is able to learn new skills using previous experience (ARC Prize Foundation, 2026). To evaluate this idea, Chollet introduced a benchmark called the Abstraction and Reasoning Corpus for Artificial General Intelligence, abbreviated as ARC-AGI (Chollet, 2019).



Figure 1—Example of an ARC-AGI problem.

Figure 1 shows an example of an ARC-AGI problem. The agent is given the input and output grids for the training examples, but for the test example it is given only the input grid and must produce up to three predictions for the output grid. If any of those predictions matches the actual test output, then the agent has successfully solved the problem. To solve the example shown above, the agent would need to recognize that all purple cells in the test should be moved below the dividing green line. The overall goal of this project was to create an agent capable of solving 96 ARC-AGI problems. Figures for all problems discussed in this report are provided in Appendix 9.3, Appendix 9.4, and Appendix 9.5.

The 96 problems were divided into three sets of 32 problems each. Each set consisted of 16 Training problems and 16 Test problems. The agent was developed and evaluated locally using the visible Training problems, while its ability to generalize was evaluated using the hidden Test problems on Gradescope. Although the Test problems were not available during development, each was designed to mimic the reasoning required by at least one corresponding Training problem. As a result, a successful approach needed to generalize beyond the visible examples rather than simply solve previously seen problems. For the remainder of this report, these three 32-problem sets will be referred to as Set B, Set C, and Set D.

These sets increase in difficulty, with Set D containing more complex problems than Set C, and Set C containing more complex problems than Set B. The problems were organized in this way so that a relatively simple agent could solve easier problems early on, while later sets required increasingly sophisticated reasoning. The development of the agent took place over four main phases, referred to as Milestone A, Milestone B, Milestone C, and Milestone D. Milestone A focused on experimenting with Set B and forming an initial plan for implementing the agent. Milestones B, C, and D then focused on progressively improving the agent’s performance on Sets B, C, and D, respectively, with the broader objective of eventually solving all 32 problems in each set.

2 AGENT DESCRIPTION

The agent solves an ARC-AGI problem by analyzing the relationship between the Training input and output grids, generating a list of candidate transformations that could explain that relationship, and then searching for a transformation or sequence of transformations that maps the input to the output. If such a sequence

is found, it is applied to the Test input to produce a prediction for the Test output. Otherwise, the agent returns the original Test input as its prediction.

A key part of this process is feature extraction. To analyze a grid, the agent first identifies objects using Connected Component Labeling with 8-connectivity, treating same-colored cells that touch orthogonally or diagonally as part of the same object. It then compares properties of the Training input and output, including object count, object shape, bounding-box size, grid size, color set, and background color. In parallel, each implemented transformation is annotated with a property profile describing which attributes it is guaranteed to preserve, which it changes, and which remain unknown. For example, the `crop` transformation changes grid size, so its `grid_size` property is marked as `changed`. Each transformation is also assigned a priority, with simpler and more computationally inexpensive transformations receiving higher priority than more complex ones.

The observed differences between the Training input and output are then compared against these transformation property profiles to produce a ranked list of candidate transformations. Candidates whose expected behavior contradicts the observed changes are treated as less likely, while those whose properties align more closely with the observations are ranked higher. After this ranking step, the agent uses Breadth-First Search to test individual transformations and short transformation sequences (programs) until it finds one that reproduces the Training output. Over time, this search process was made more robust by adding safeguards such as a maximum search depth and fixes that prevent unnecessary continuation after a goal state has already been found.

Although the overall pipeline remained consistent throughout the project, the agent became more capable through the addition of new transformations and supporting helper methods. In Milestone C, the Milestone B framework was extended with transformations for more complex patterns, such as connecting matching pixels across the grid, moving pixels relative to anchors, and handling grids with multiple divider lines. In Milestone D, the same general pipeline was retained, but additional transformations and helper methods were introduced for more specialized tasks, such as identifying object gaps, aligning objects with those gaps, and filling the centers of cycles. As a result, the agent improved less through changes to its overall architecture and more through the expansion and refinement of the transformations and analysis methods within that architecture.

3 AGENT PERFORMANCE

The agent performed well overall across Sets B, C, and D. On Gradescope, it achieved a perfect score of 16/16 on the Training problems for all three sets. On the hidden Test problems, it achieved 15/16 for Set B, 15/16 for Set C, and 11/16 for Set D.

The hidden Test failures were relatively limited in Sets B and C, but more numerous in Set D. In Set B, the agent failed only problem [28e73c20](#). In Set C, it failed only problem [7b6016b9](#). In Set D, it failed problems [2546ccf6](#), [4b6b68e5](#), [60a26a3e](#), [992798f6](#), and [e9b4f6fc](#).

Although the agent achieved perfect performance on the visible Training problems for all three sets, the hidden Test results provide a better measure of generalization. The relatively small gap between Training and Test performance in Sets B and C suggests that the implemented transformations generalized well to unseen variants of those problems. In contrast, the larger gap in Set D indicates that the most difficult problems exposed limitations in the generality of some transformations and the agent’s ability to handle more complex reasoning patterns.

A closer look at the local runtime results, shown below in Figure 2, shows a similar pattern. In Set B, the shortest runtime was for problem [6150a2bd](#) at 0.0001406 seconds, while the longest was for [ce4f8723](#) at 0.0166452 seconds. In Set C, the shortest runtime was for [bbc9ae5d](#) at 0.0003160 seconds, while the longest was for [b2862040](#) at 0.0133490 seconds. In Set D, the shortest runtime was for [c1990cce](#) at 0.0002297 seconds, while the longest was for [4b6b68e5](#) at 0.0043151 seconds. Overall, all problems were still solved within milliseconds locally, but the more difficult problems generally required more time because they involved more complex analysis and lower priority transformations.

In terms of total runtime on Gradescope, the agent completed Set B in 2.067 seconds, Set C in 8.898 seconds, and Set D in 12.195 seconds. This increase in runtime is consistent with the increasing difficulty of the three sets. As the problems became more complex, the agent needed to consider more specialized transformations and more difficult transformation sequences, which increased the amount of time required to solve each set.

Additional local runtime details are provided in Appendix 9.1. Complete Grade-scope results are shown in Appendix 9.2.

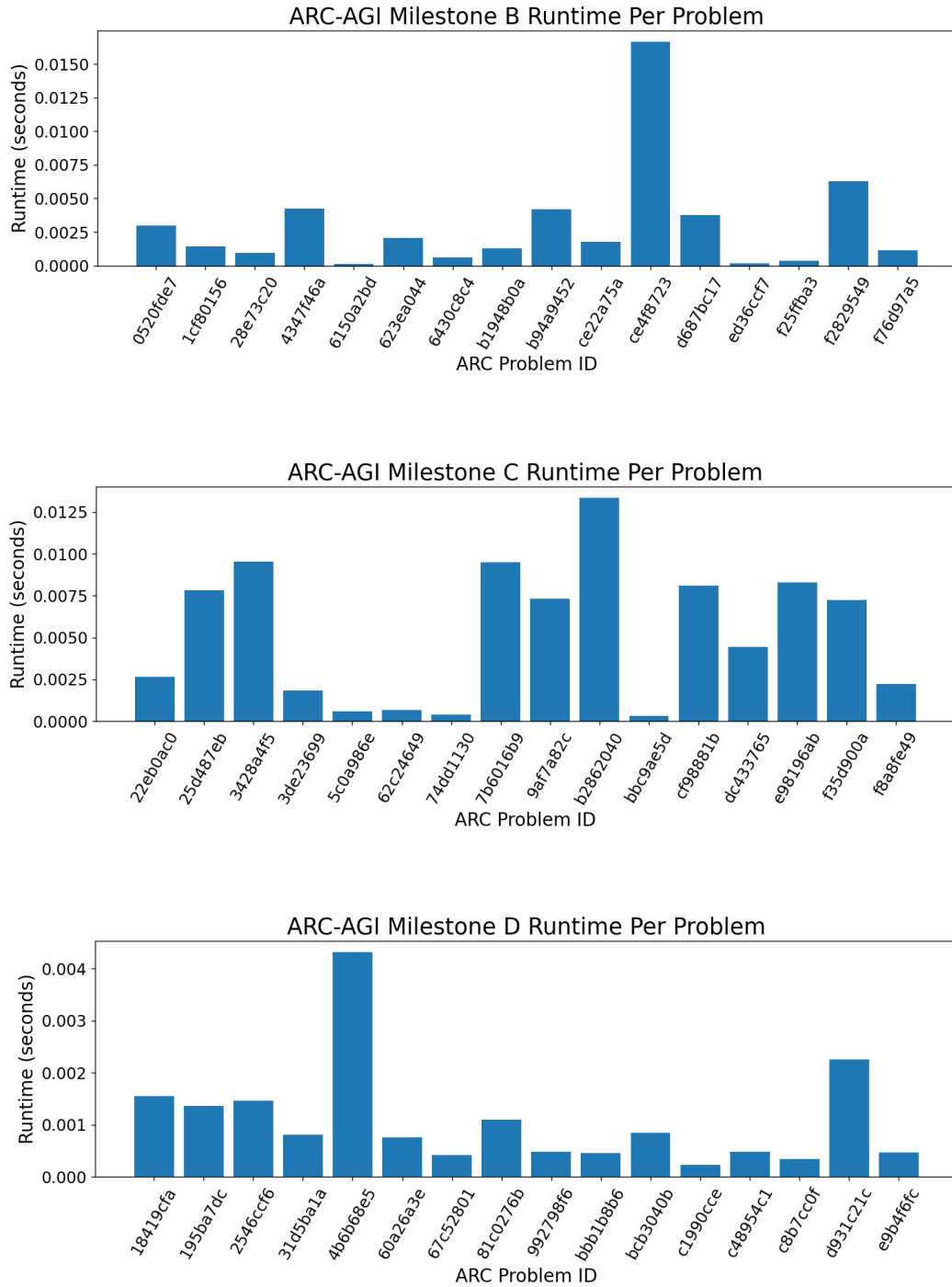


Figure 2—Local runtime per problem for the Training sets of Milestones B, C, and D.

4 AGENT SUCCESSES

The agent was able to solve both the Training and Test versions of problem [1cf80156](#) from Set B. This problem is one of the simpler success cases because it contains only a single object against a black background, so the required transformation is simply to remove all rows and columns that do not contain any part of the object.

The agent was able to solve both the Training and Test versions of problem [f8a8fe49](#) from Set C. This problem is more difficult than [1cf80156](#) because the agent must first identify the bracket-shaped objects, then determine how the position of the inner objects should be mirrored relative to the bracket it is closest to. Unlike the previous problem, this one requires the agent to keep track of more than one object and reason about their spatial relationship. The transformation is therefore not just a simple crop or direct grid-level change, but a structured repositioning of the inner objects based on where the surrounding brackets are located.

The agent was able to solve both the Training and Test versions of problem [67c52801](#) from Set D. This problem is more difficult than the previous two because the agent must not only determine which objects need to be moved, but also keep track of multiple objects and multiple gaps at the same time so that each gap is filled by an appropriate object. In addition, the agent must determine whether an object needs to be rotated before it can fill a gap, and then line the object up with the correct rows or columns of that gap. This makes the transformation more complex since the agent must reason about several object-gap relationships at once rather than handling a single object or a single movement.

5 AGENT STRUGGLES

The agent was not able to solve the Test version of problem [28e73c20](#) from Set B. This problem starts with an empty grid, and the agent must draw a spiral shape onto the grid. While this sounds simple, the agent must be able to draw spirals of different sizes on grids of different dimensions while still preserving the correct direction, spacing, and stopping point of the spiral.

The agent was not able to solve the Test version of problem [7b6016b9](#) from Set C. This problem requires the agent to color all cells inside the object one color while coloring all cells outside the object another color. Although the overall rule is

clear, the difficulty comes from correctly identifying the boundary of the object and distinguishing its interior from the surrounding exterior across a large and irregular grid.

The agent was not able to solve the Test version of problem [2546ccf6](#) from Set D. This problem requires the agent to complete each partially formed object by determining which section of the grid the missing part belongs in and then drawing the missing part correctly within the appropriate section. Compared to the earlier failures, this problem is more difficult because the agent must reason about multiple grid sections at once and determine how the visible pieces should be mirrored or extended to complete the full object.

The most likely reason the agent failed these Test problems is that the current implementation is not general enough. Multiple attempts were made to make the agent more general so that it could pass these cases; however, those changes did not solve the issue.

6 AGENT DEVELOPMENT

The overall development of the agent was iterative rather than based on one fixed approach from the beginning. In Milestone A, the original idea was to build a set of basic transformation functions and guide the agent using a decision tree style set of heuristics. The purpose of the decision tree was not to directly choose a solution, but to narrow the range of possible transformations based on high-level properties of the Training input and output, such as whether the grid size changed, whether pixels were added or removed, or whether colors were modified. This initial idea was a useful starting point because it reflected how a human might reason through the problems, but it became clear early on that a fixed decision tree would be too rigid for more varied ARC-AGI problems and would be difficult to scale as the transformations became more specific and complex.

As development continued, the agent moved away from the earlier decision tree idea and toward a transformation-based search approach. Rather than trying to hard-code a sequence of branching rules for every kind of problem, the focus shifted to implementing a growing set of transformations that the agent could apply to the Training input grids. A major development during Milestone B was the introduction of feature-based transformation analysis and scoring. Instead of

relying on a heuristic tree to decide what to do next, the agent began analyzing the Training input and output, extracting object- and grid-level features, and comparing those observations against the expected properties of each candidate transformation. This made the approach more flexible, since the same general framework could still be used even as new transformations were added later in the project.

From that point on, the overall direction of development remained the same, but the agent became more capable through repeated refinement rather than a complete redesign.

7 AGENT-HUMAN COMPARISON

In some ways, the agent's approach is similar to human's, but it is also different in other ways. Similar to a human, the agent starts by comparing the Training input and output grids, looking for any high-level patterns that are preserved or removed. These patterns include things such as if the grid size changed, if the same object appears in both grids, or if the color sets of the grids are different. In this sense, the agent is similar to a human.

However, the agent is still less flexible than a human. A human can often look at a problem and quickly recognize the most meaningful structure in the grid, even when the required transformation is unusual or highly specific. In contrast, the agent must pull from its available library of transformations in order to solve the problem. If a problem contains a very specific transformation that is not in its library, the agent is unable to provide a prediction that would match the solution.

8 CONCLUSION

The goal of this project was to create an agent capable of solving 96 ARC-AGI problems. To accomplish this goal, the agent was developed iteratively so that it could not only determine which transformations were needed to solve a problem, but also how to apply those transformations correctly. The agent was evaluated on three increasingly complex sets of ARC-AGI problems. In Set B, it solved 31/32 problems. In Set C, it solved 31/32 problems. In Set D, it solved 27/32 problems. Overall, the agent solved 89/96 ARC-AGI problems, corresponding to success rate of 92.71%, which is relatively strong performance across the full benchmark.

REFERENCES

- [1] ARC Prize Foundation (2026). *What is ARC-AGI?* <https://arcprize.org/arc-agi>. Accessed 2026-04-16.
- [2] Chollet, François (2019). “On the Measure of Intelligence”. In: *arXiv preprint arXiv:1911.01547*.

9 APPENDIX

9.1 Local Runtimes

Milestone B		Milestone C		Milestone D	
Problem	Runtime (s)	Problem	Runtime (s)	Problem	Runtime (s)
0520fde7	0.0030057	22eboaco	0.0026651	18419cfa	0.0015518
1cf80156	0.0014553	25d487eb	0.0078377	195ba7dc	0.0013553
28e73c20	0.0009291	3428a4f5	0.0095521	2546ccf6	0.0014582
4347f46a	0.0042591	3de23699	0.0018506	31d5ba1a	0.0008051
6150a2bd	0.0001406	5coa986e	0.0005755	4b6b68e5	0.0043151
623ea044	0.0020568	62c24649	0.0006811	60a26a3e	0.0007505
6430c8c4	0.0006075	74dd1130	0.0003878	67c52801	0.0004169
b1948boa	0.0012973	7b6016b9	0.0094805	81c0276b	0.0010948
b94a9452	0.0041879	9af7a82c	0.0073177	992798f6	0.0004807
ce22a75a	0.0017551	b2862040	0.0133490	bbb1b8b6	0.0004525
ce4f8723	0.0166452	bbc9ae5d	0.0003160	bcb3040b	0.0008473
d687bc17	0.0037630	cf98881b	0.0080922	c199occe	0.0002297
ed36ccf7	0.0001624	dc433765	0.0044319	c48954c1	0.0004761
f25ffba3	0.0003811	e98196ab	0.0082788	c8b7ccof	0.0003416
f2829549	0.0062996	f35d900a	0.0072499	d931c21c	0.0022483
f76d97a5	0.0011590	f8a8fe49	0.0022268	egb4f6fc	0.0004642

Table 1—Runtime for ARC problems by milestone.

9.2 Gradescope Results

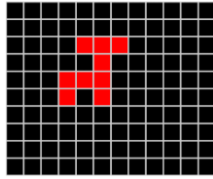
Milestone B			Milestone C			Milestone D		
Problem	Train	Test	Problem	Train	Test	Problem	Train	Test
0520fde7	Pass	Pass	22eboaco	Pass	Pass	18419cfa	Pass	Pass
1cf80156	Pass	Pass	25d487eb	Pass	Pass	195ba7dc	Pass	Pass
28e73c20	Pass	Fail	3428a4f5	Pass	Pass	2546ccf6	Pass	Fail
4347f46a	Pass	Pass	3de23699	Pass	Pass	31d5ba1a	Pass	Pass
6150a2bd	Pass	Pass	5coa986e	Pass	Pass	4b6b68e5	Pass	Fail
623ea044	Pass	Pass	62c24649	Pass	Pass	60a26a3e	Pass	Fail
6430c8c4	Pass	Pass	74dd1130	Pass	Pass	67c52801	Pass	Pass
b1948boa	Pass	Pass	7b6016b9	Pass	Fail	81c0276b	Pass	Pass
b94a9452	Pass	Pass	9af7a82c	Pass	Pass	992798f6	Pass	Fail
ce22a75a	Pass	Pass	b2862040	Pass	Pass	bbb1b8b6	Pass	Pass
ce4f8723	Pass	Pass	bbc9ae5d	Pass	Pass	bc3040b	Pass	Pass
d687bc17	Pass	Pass	cf98881b	Pass	Pass	c1990cce	Pass	Pass
ed36ccf7	Pass	Pass	dc433765	Pass	Pass	c48954c1	Pass	Pass
f25ffba3	Pass	Pass	e98196ab	Pass	Pass	c8b7ccof	Pass	Pass
f2829549	Pass	Pass	f35d900a	Pass	Pass	d931c21c	Pass	Pass
f76d97a5	Pass	Pass	f8a8fe49	Pass	Pass	e9b4f6fc	Pass	Fail
Passed	16/16	15/16	Passed	16/16	15/16	Passed	16/16	11/16
Runtime (s)	2.067		Runtime (s)	8.898		Runtime (s)	12.195	

Table 2—Gradescope results by milestone.

9.3 Milestone B Problems

1cf80156

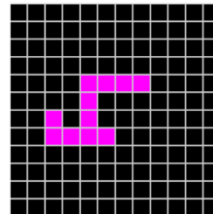
Train Data In:



Train Data Out:



Test Data In:



Test Data Out:

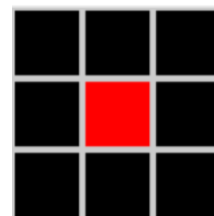


0520fde7

Train Data In:



Train Data Out:



Test Data In:

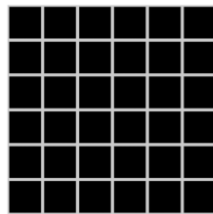


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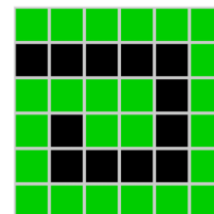


28e73c20

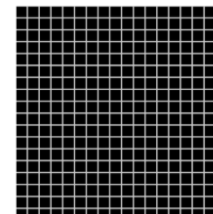
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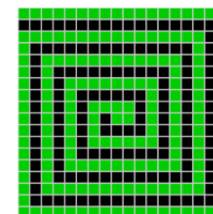
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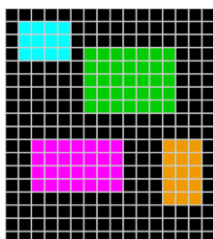


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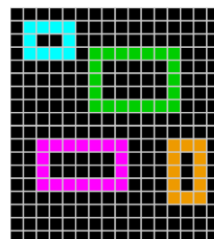


4347f46a

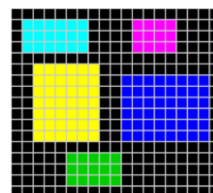
Train Data In:



Train Data Out:



Test Data In:



Test Data Out:



6150a2bd

Train Data In:



Train Data Out:



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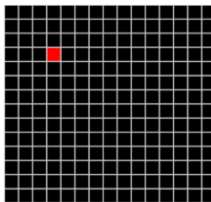


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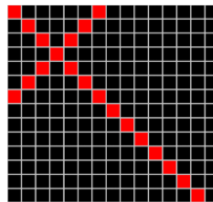


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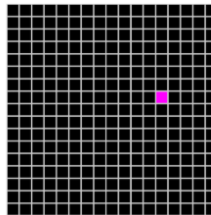
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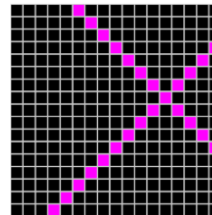
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Test Data Out:

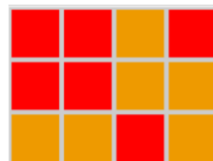


b1948boa

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Train Data Out:



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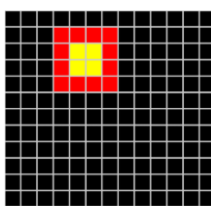


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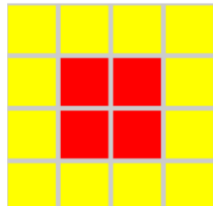


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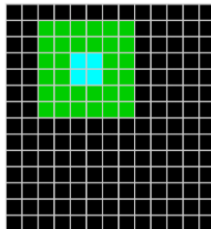
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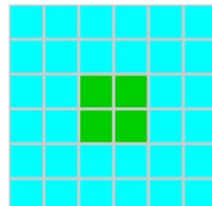
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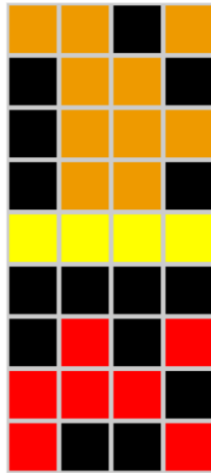


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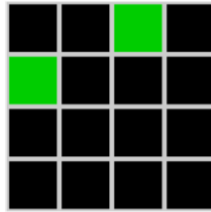


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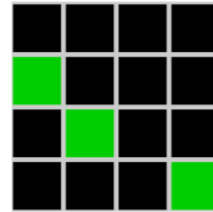
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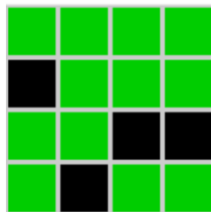


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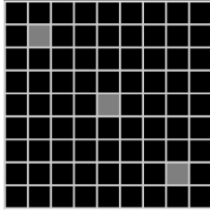


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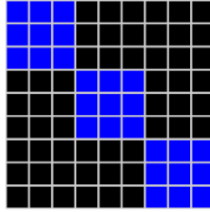


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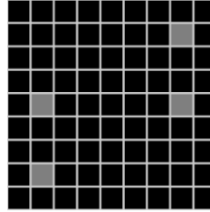
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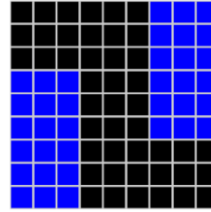
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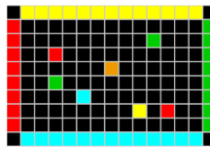


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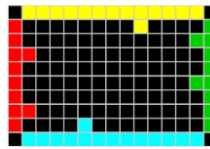


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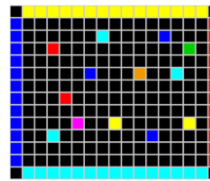
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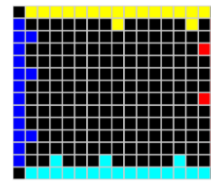
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Test Data Out:



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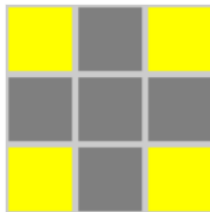


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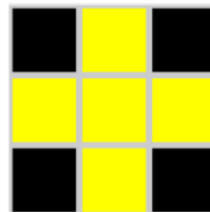


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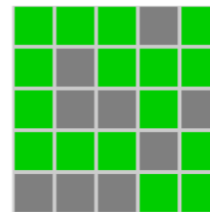
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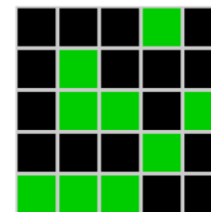
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Test Data Out:



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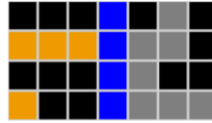
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Train Data Out:



Test Data In:



Test Data Out:



f25ffb3

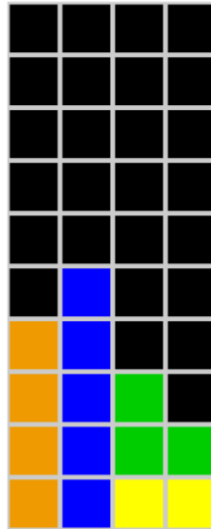
Train Data In:



Train Data Out:



Test Data In:



Test Data Out:



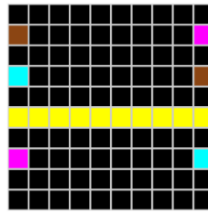
9.4 Milestone C Problems

22eboaco

Train Data In:



Train Data Out:



Test Data In:

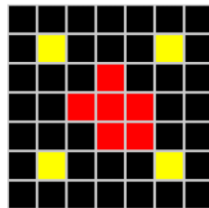


Test Data Out:

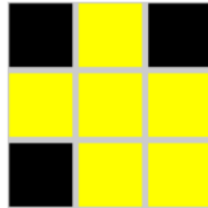


3de23699

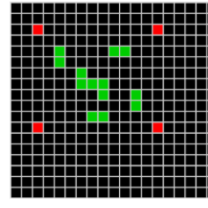
Train Data In:



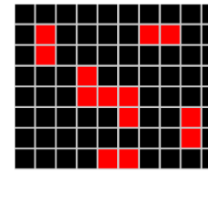
Train Data Out:



Test Data In:

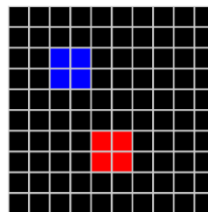


Test Data Out:

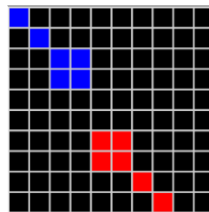


5co986e

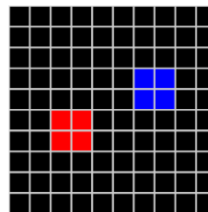
Train Data In:



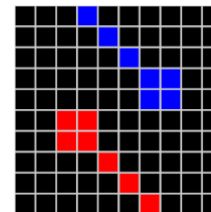
Train Data Out:



Test Data In:



Test Data Out:



62c24649

Train Data In:



Train Data Out:



Test Data In:

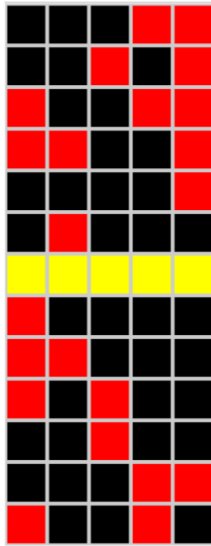


Test Data Out:

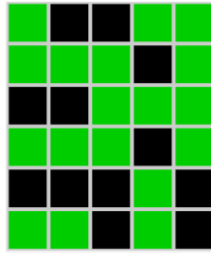


3428a4f5

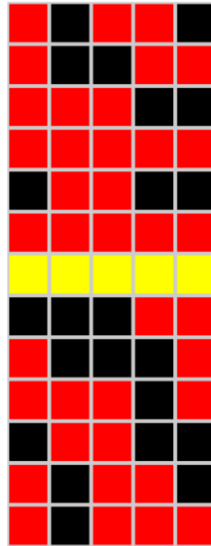
Train Data In:



Train Data Out:



Test Data In:

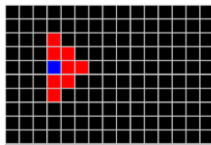


Test Data Out:

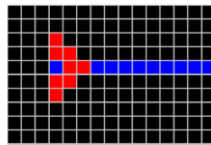


25d487eb

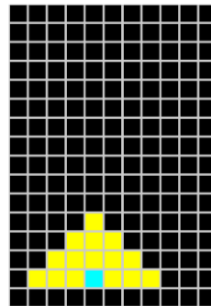
Train Data In:



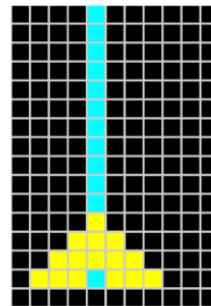
Train Data Out:



Test Data In:

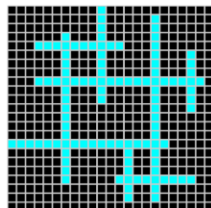


Test Data Out:

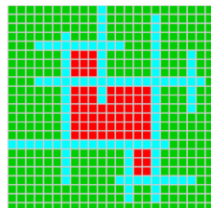


7b6016b9

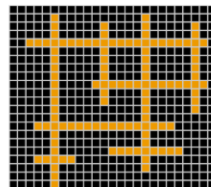
Train Data In:



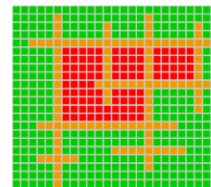
Train Data Out:



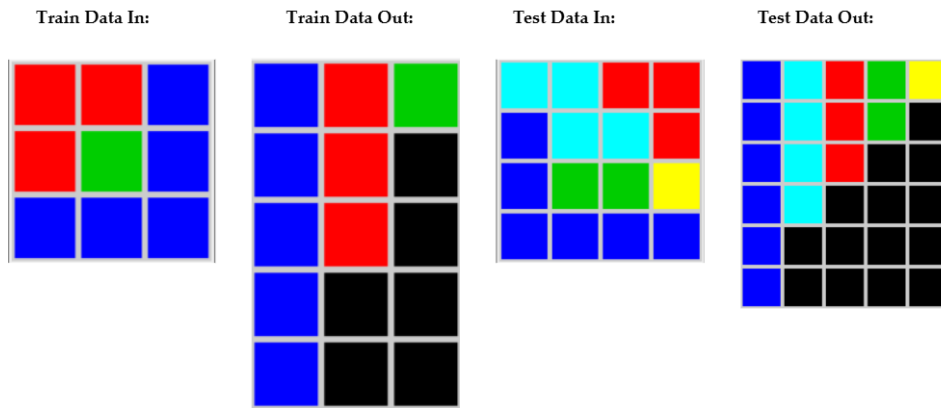
Test Data In:



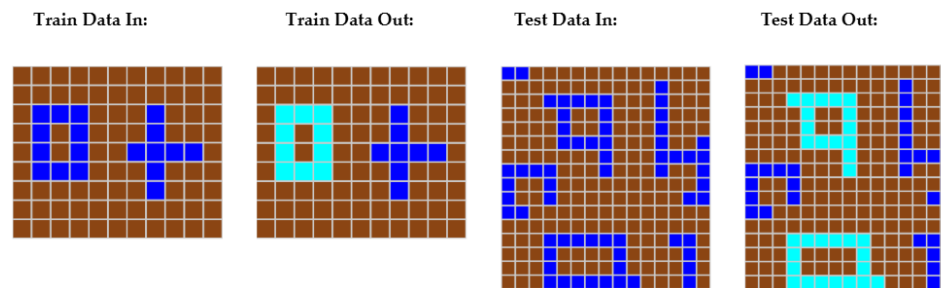
Test Data Out:



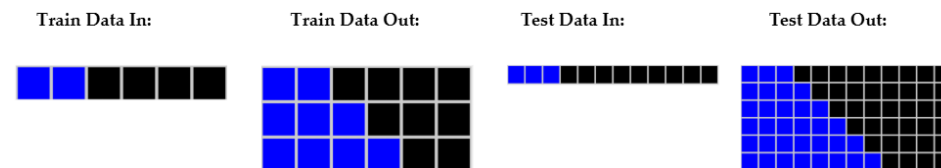
9af7a82c



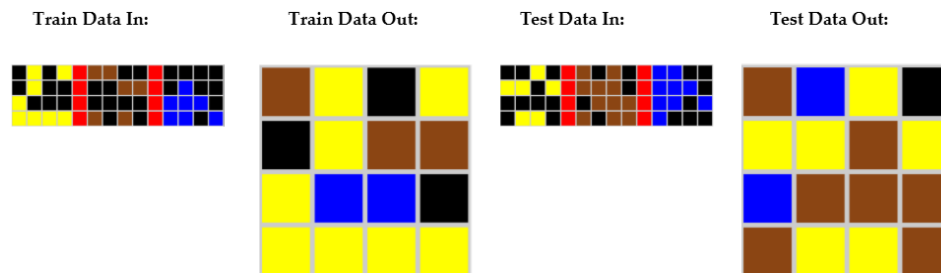
b2862040



bbc9ae5d

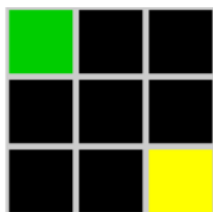


cf98881b

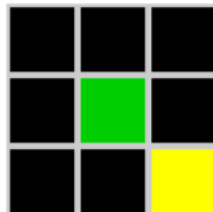


dc433765

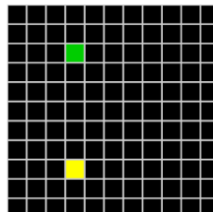
Train Data In:



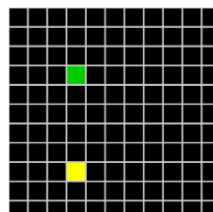
Train Data Out:



Test Data In:

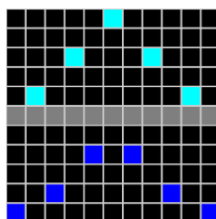


Test Data Out:

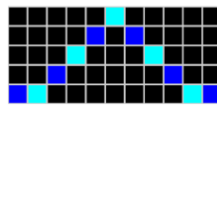


e98196ab

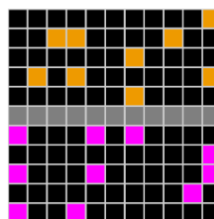
Train Data In:



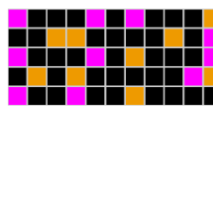
Train Data Out:



Test Data In:

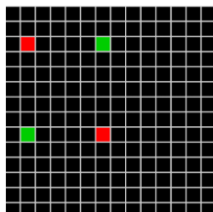


Test Data Out:



f35d900a

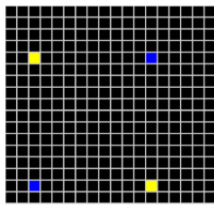
Train Data In:



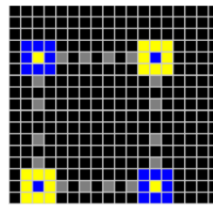
Train Data Out:



Test Data In:

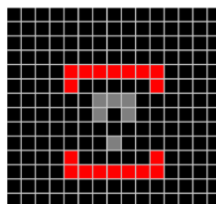


Test Data Out:

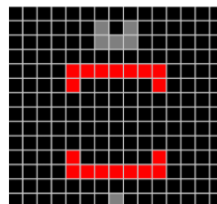


f8a8fe49

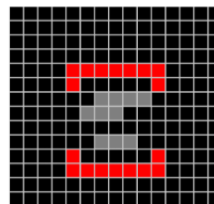
Train Data In:



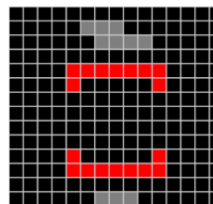
Train Data Out:



Test Data In:



Test Data Out:



74dd1130

Train Data In:



Train Data Out:



Test Data In:



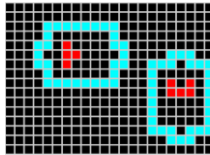
Test Data Out:



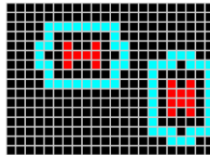
9.5 Milestone D Problems

18419cfa

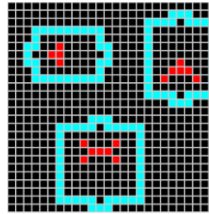
Train Data In:



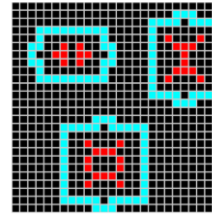
Train Data Out:



Test Data In:



Test Data Out:

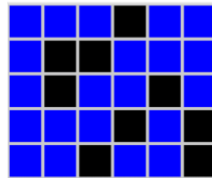


195ba7dc

Train Data In:



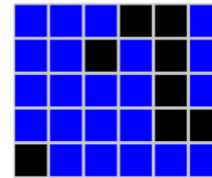
Train Data Out:



Test Data In:

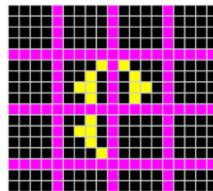


Test Data Out:

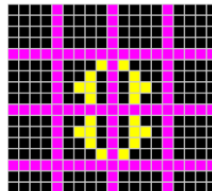


2546ccf6

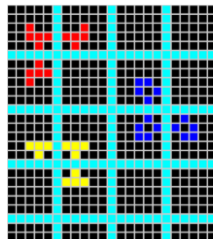
Train Data In:



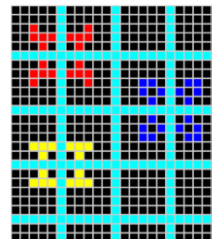
Train Data Out:



Test Data In:



Test Data Out:



31d5ba1a

Train Data In:



Train Data Out:



Test Data In:

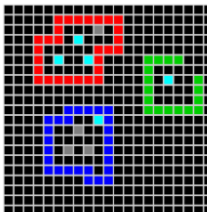


Test Data Out:

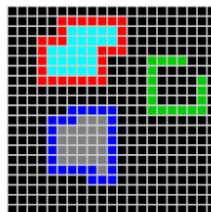


4b6b68e5

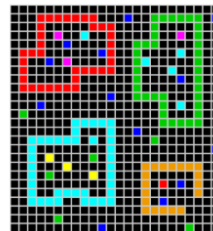
Train Data In:



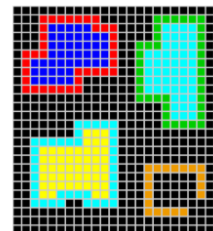
Train Data Out:



Test Data In:

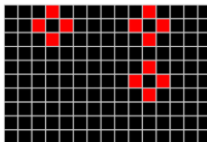


Test Data Out:

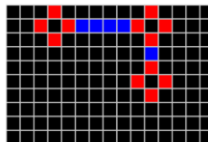


60a26a3e

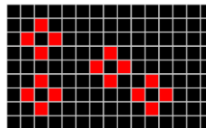
Train Data In:



Train Data Out:



Test Data In:



Test Data Out:

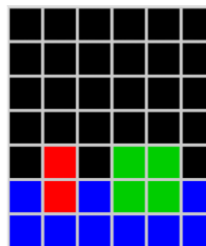


67c52801

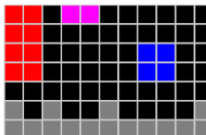
Train Data In:



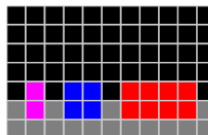
Train Data Out:



Test Data In:

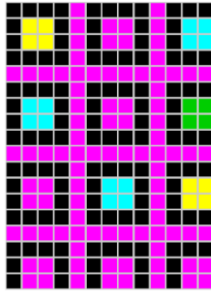


Test Data Out:



81c0276b

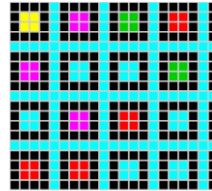
Train Data In:



Train Data Out:



Test Data In:

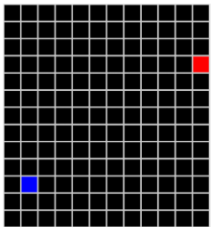


Test Data Out:

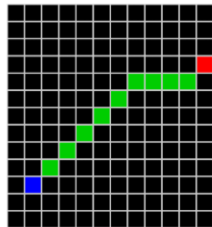


992798f6

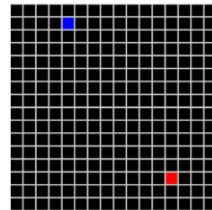
Train Data In:



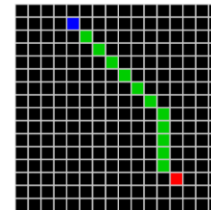
Train Data Out:



Test Data In:

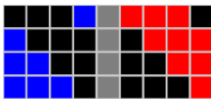


Test Data Out:

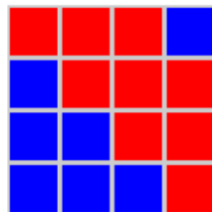


bbb1b8b6

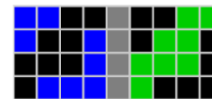
Train Data In:



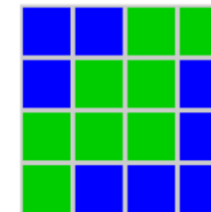
Train Data Out:



Test Data In:

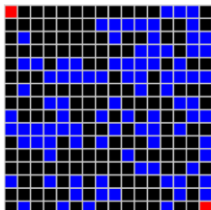


Test Data Out:

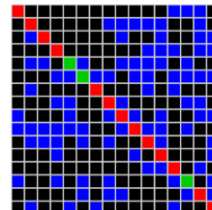


bcb3040b

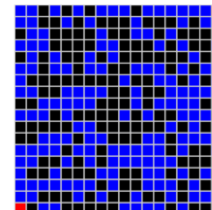
Train Data In:



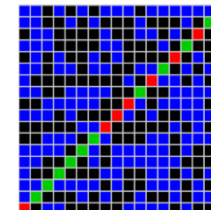
Train Data Out:



Test Data In:



Test Data Out:

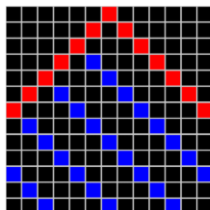


c1990cce

Train Data In:



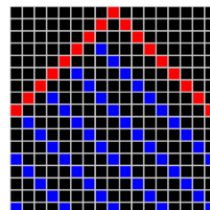
Train Data Out:



Test Data In:



Test Data Out:



c48954c1

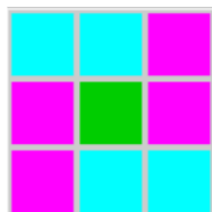
Train Data In:



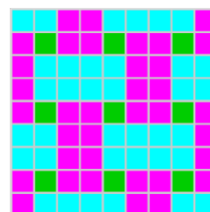
Train Data Out:



Test Data In:

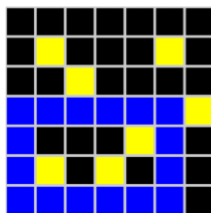


Test Data Out:

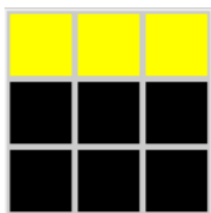


c8b7ccof

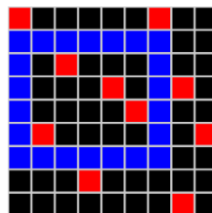
Train Data In:



Train Data Out:



Test Data In:

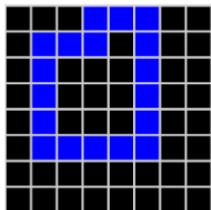


Test Data Out:



d931c21c

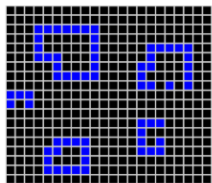
Train Data In:



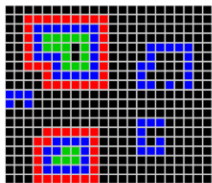
Train Data Out:



Test Data In:

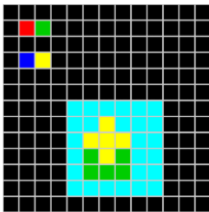


Test Data Out:

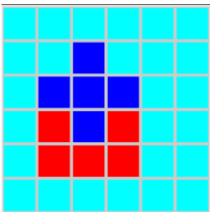


e9b4f6fc

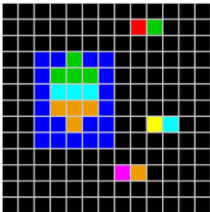
Train Data In:



Train Data Out:



Test Data In:



Test Data Out:

